

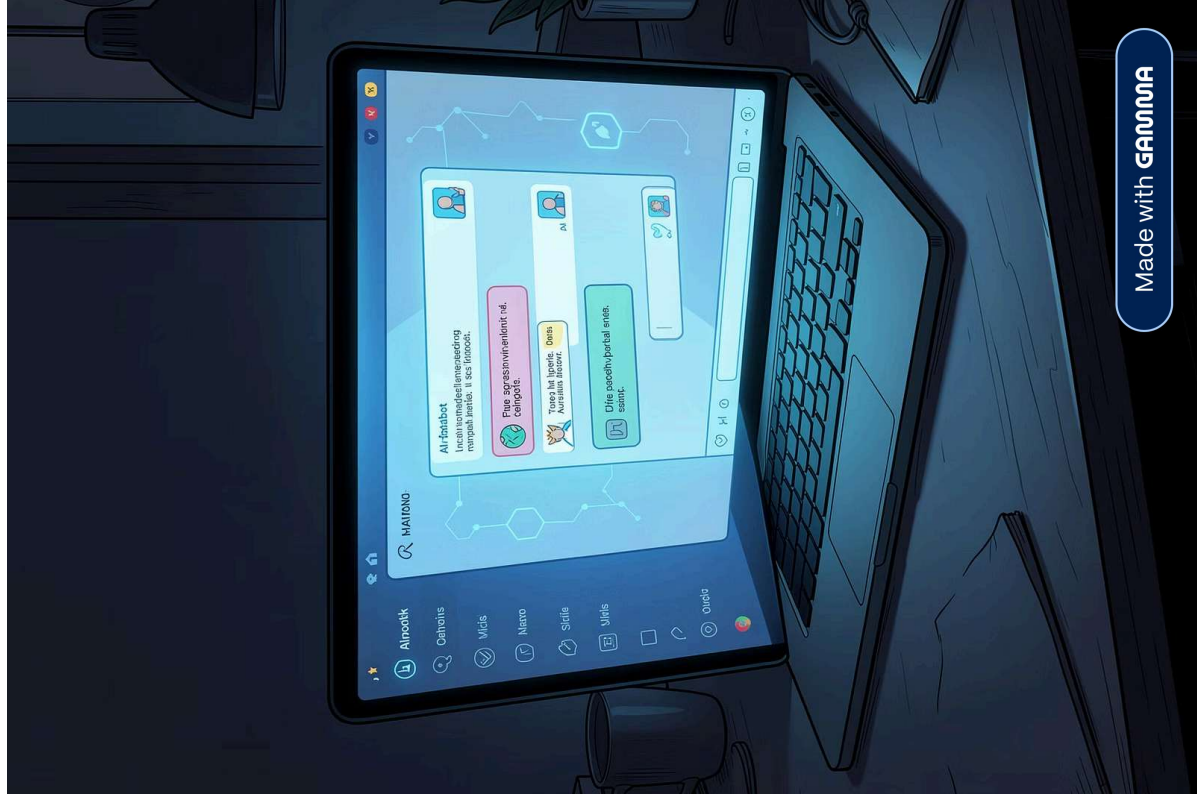
TinyLlama Chatbot

An **offline AI chatbot** built with **Streamlit** and **Ollama** — interact with a local LLM with no API key and no internet connection required. 🚀

OFFLINE AI

TINYLLAMA

STREAMLIT





Why Run AI Locally?



Privacy First

Your conversations never leave your machine. No data sent to external servers.



Zero Latency

No network calls. Responses come directly from your local Ollama instance.



No API Key

Completely free to run. No subscriptions, no rate limits, no credentials.



Beginner-Friendly

Minimal setup. Just Python, Ollama, and a few commands to get started.

Project Features



Fully Offline

Runs 100% on your machine — no internet required after setup.



Lightweight Model

TinyLlama is small and fast, ideal for local inference on modest hardware.



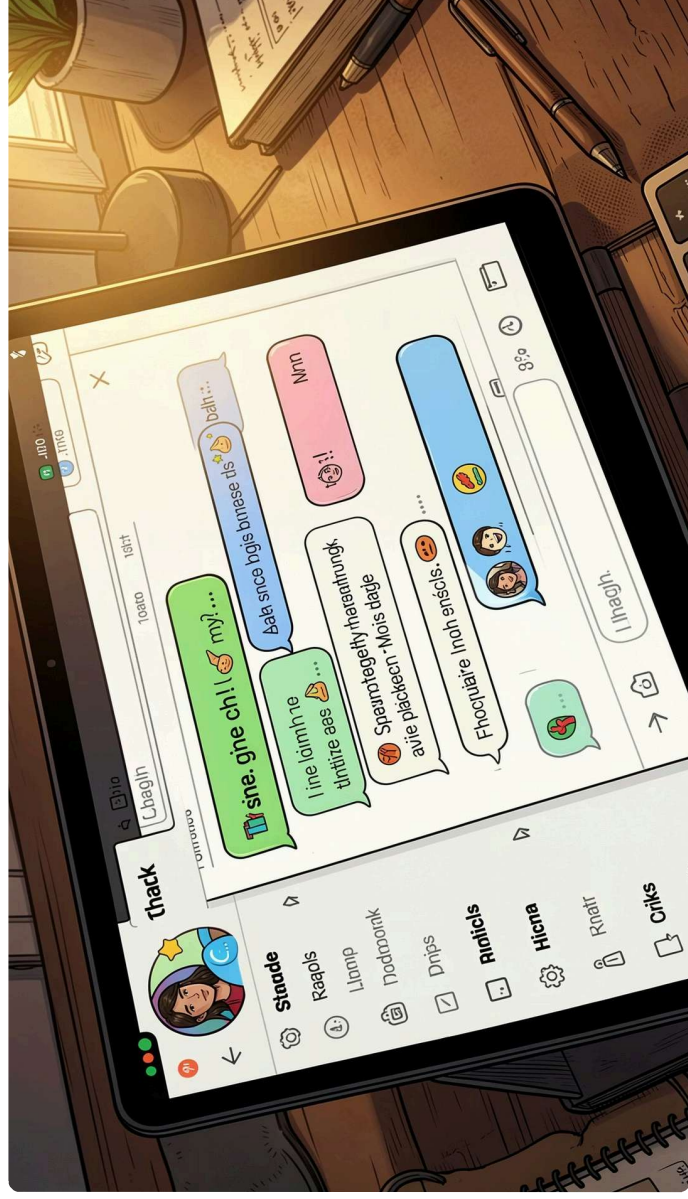
Interactive UI

Streamlit provides a clean, responsive chat interface in the browser.



No Credentials

No `.env` file, no API keys — just run and chat.



Tech Stack



Python

Core language powering the application logic and Ollama integration.



Ollama

Lightweight runtime for running large language models locally.



Streamlit

Turns Python scripts into interactive web apps — no frontend skills needed.



TinyLlama

A compact 1.1B parameter LLM optimized for speed and low resource usage.

Project Structure

File Overview

The project is intentionally minimal — just three files keep everything organized and easy to understand.

Directory Tree

```
TinyLlama Project/  
├── TinyLlamaAi/  
│   ├── TinyLlama.py  
│   ├── .env  
│   └── README.md
```

 Keep `TinyLlama.py` inside the `TinyLlamaAi/` folder when running the app.

`TinyLlama.py`

Main Streamlit app — handles UI and Ollama API calls.

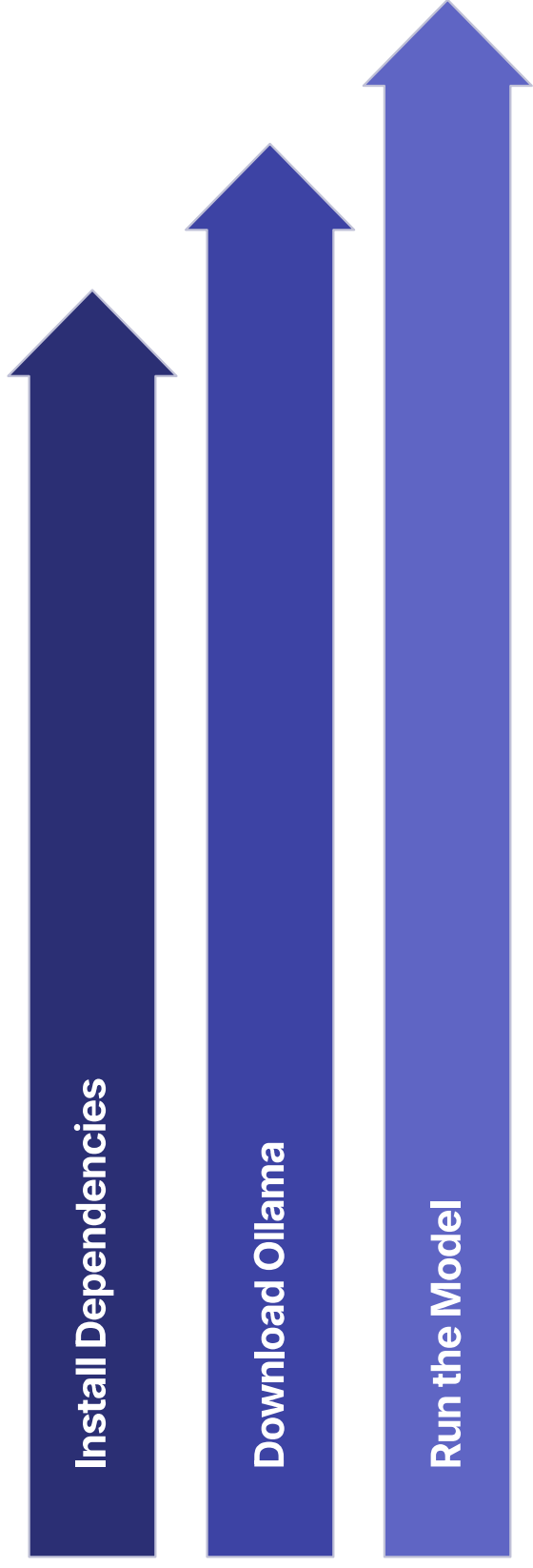
`.env`

Not required for offline use. Included for future extensibility.

`README.md`

Setup instructions and project documentation.

Installation & Ollama Setup



Follow these three steps in order. Make sure to keep the Ollama terminal running — do not close it while using the chatbot.

⚠ Keep the Ollama terminal open at all times.
Closing it will break the connection to the chatbot.

Quick Commands

```
pip install streamlit requests  
ollama run tinyllama
```

How to Run the Project

1

Start Ollama

Open Terminal 1 and run ollama run tinylama. Leave this running.

2

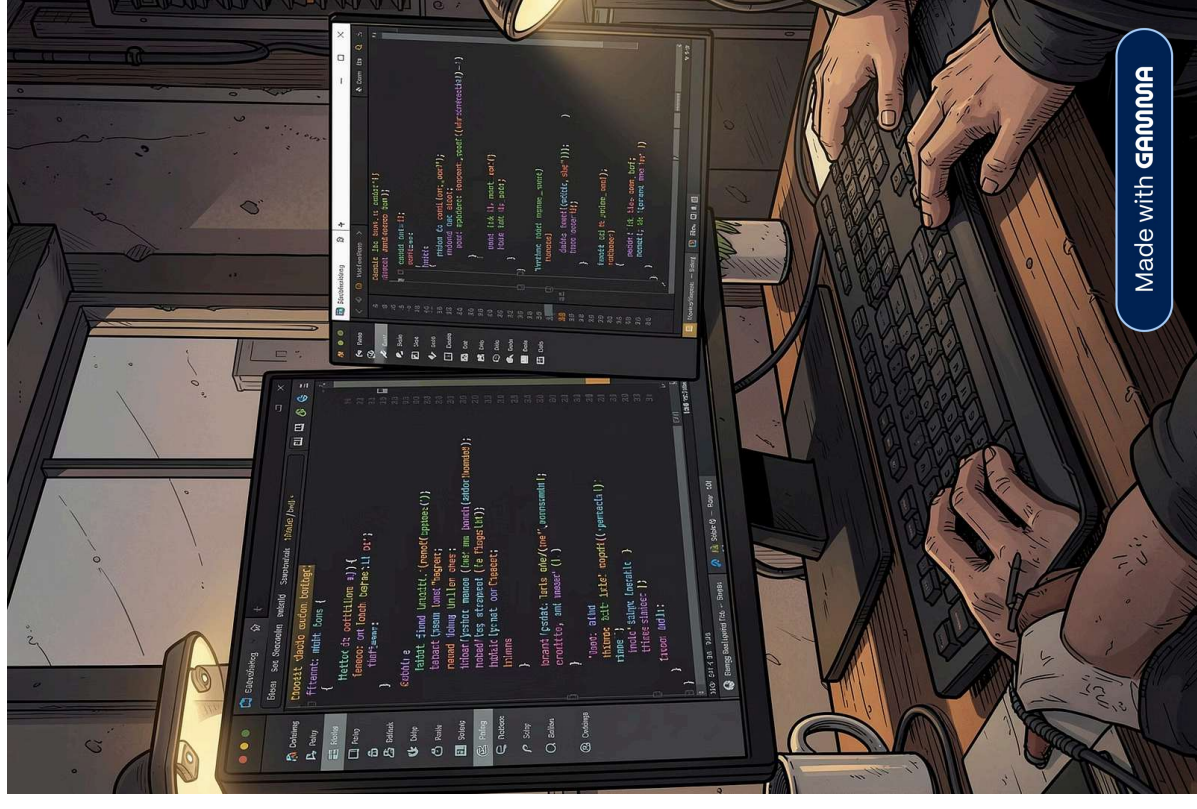
Launch Streamlit

Open Terminal 2, navigate with cd TinyLlamaAi, then run streamlit run TinyLlama.py.

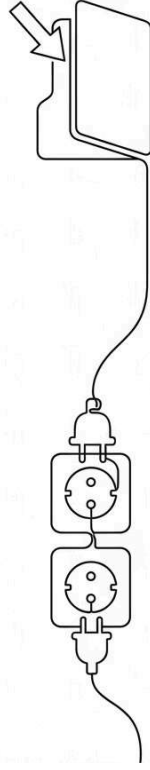
3

Open Browser

Go to <http://localhost:8501> — your chatbot UI will load automatically.



Troubleshooting



Common Errors & Fixes: Connected.

Ensure Ollama is running with

ollama run tinyllama

Common Errors & Fixes: File Not Found

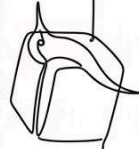
Verify you are in the correct folder with

cd TinyLlamaAi



Common Errors & Fixes: Module Not Found

Install missing packages with
pip install streamlit maailit requests



Common Pitfalls

- **Connection Refused** — Ollama isn't running. Start it first.
- **File Not Found** — You're in the wrong directory. Use `cd TinyLlamaAi`.
- **Module Not Found** — Dependencies missing. Run `pip install streamlit requests`.

📌 Always start Ollama **before** launching Streamlit. Order matters!

What's Next?

The project is a solid foundation. Here's what's on the roadmap for future iterations.



Voice Chatbot

Add speech-to-text and text-to-speech for hands-free interaction.



Multi-Language

Support for Marathi and English to broaden accessibility.



Better Models

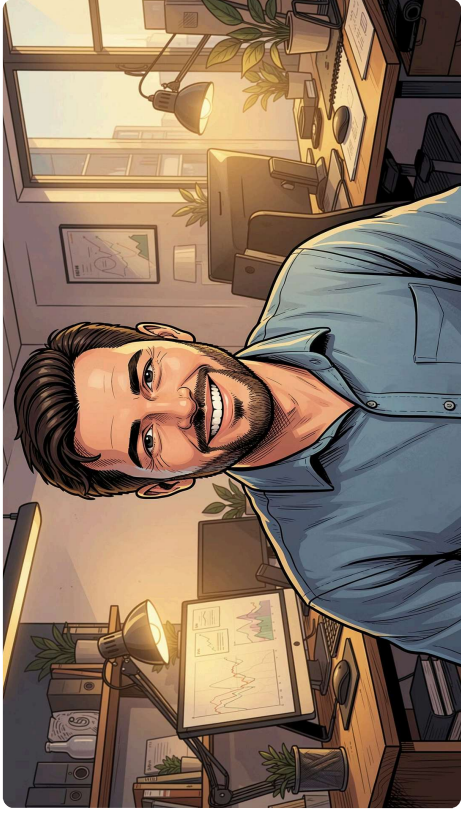
Swap TinyLlama for Phi-3 or Mistral for improved responses.



UI Enhancements

Polish the Streamlit interface with custom themes and layouts.

Author & Support



Built by Pallavi Khadse

This project demonstrates how accessible local AI can be — no cloud, no cost, no complexity.

Offline AI is not just a project — it's a statement about privacy and accessibility. 🚀

★ Star on GitHub

If this project helped you, show your support with a star on the repository.

👉 Contribute

Fork, improve, and submit a PR. Every contribution makes it better.